Young Hyun Kim

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RESEARCH INTERESTS

Human cardiac tissue engineering and organoid models, chamber-specific electromechanical function, mechanobiology, and quantitative phenotyping of hiPSC-derived cardiac systems.

EDUCATION

Sejong University Seoul, Republic of Korea

M.S. in Integrative Biological Sciences and Industry **Aug 2026**

GPA: 4.07/4.5

Thesis: *Generation of Chamber-Specific 3D Cardiac Organoids and an Integrated Electro-Mechanical Optical Analysis Platform*.

B.S. in Integrative Biological Sciences and Industry **Aug 2023**

GPA: 3.50/4.5

PUBLICATIONS

1. **Kim, Y. H.**, Son, Y. H., Choi, Y., Kim, M. S., Park, S. J., & Lee, K. Y. (2026). Advanced in vitro cardiac models for drug evaluation: Integration of organoids, engineered tissues, and microphysiological systems. *Microsystems & Nanoengineering*, 12(1), 162. **(Review)**
2. Kim, M. S., **Kim, Y. H.**, Choi, Y., & Lee, K. Y. (2025). Biohybrid actuators in robotics: recent trends and future perspectives of skeletal and cardiac muscle integration. *npj Robotics*, 3(1), 37. **(Review)**
3. Kim, Y. S., Lee, M., Cheon, Y. J., Choi, H. S., Choi, B. J., **Kim, Y. H.**, Lee, K. Y., & Ryu, C. J. (2026). Targeting CD147 or CD98hc is sufficient to collapse the CD147–CD98hc–MCT1/MCT4 module and restrain hepatocellular carcinoma growth. *Cancer Research Communications*. **(*In preparation*)**

RESEARCH EXPERIENCE

Biological Systems Engineering Laboratory, Sejong University Seoul, Korea

Dept. of Integrative Biological Sciences and Industry · Prof. Keel Yong Lee

Research Assistant Sep 2026 – Present

Integrated Electromechanical Stimulation and Incubation System

- Developing a new tissue stimulation system combining electrical pacing and mechanical stretch with integrated incubation capabilities; defining experimental requirements and coordinating fabrication.

Dual-Camera Fluorescence/Brightfield Imaging System

- Extending my master's work on calcium and motion analysis by developing a dual-camera system with a dichroic beam splitter to enable simultaneous Fluo-4 calcium-transient imaging and brightfield recording of contractile motion in cardiac organoids.

Graduate Research Assistant Sep 2023 – Aug 2026

Thesis Project: Chamber-Specific Cardiac Organoids and Integrated Functional Analysis

- Generated atrial- and ventricular-like 3D cardiac organoids using a biphasic 2D-to-3D differentiation workflow with stage-specific Wnt and retinoic-acid modulation.
- Assessed lineage specification by qRT-PCR and cardiomyocyte purity by cTnT flow cytometry in monolayer cultures; characterized 3D organoid structure and marker expression by immunofluorescence.
- Developed iCAMAnalyzer, a Python-based platform integrating calcium-transient analysis, calcium-based activation mapping, and label-free motion analysis, and applied it to compare atrial- and ventricular-like cardiac organoids.
- Benchmarked activation mapping against synthetic ground-truth videos (CV recovery $R^2 > 0.99$ for tested planar and radial waves) and assessed sensitivity to frame rate, noise, and spatial resolution.
- Identified faster, shorter calcium transients and higher spontaneous beating rates in atrial-like versus ventricular-like organoids; calcium-derived propagation velocities were comparable.
- Developed a MATLAB morphology analyzer and compared diameter measurements with manual ImageJ analysis ($R^2 = 0.973$; $n = 27$).
- Demonstrated parallel optical analysis on a nine-organoid micro-pillar array and tested motion analysis on organoid samples provided by Konkuk University researchers.

Additional Research & Collaborations

- Investigated spinner-flask culture conditions for scaling hiPSC-derived cardiomyocyte production and organoid fabrication.
- Differentiated and supplied hiPSC-derived cardiomyocytes and jointly performed flow cytometry characterization for a co-authored study in Cancer Research Communications.

Undergraduate Research Assistant Oct 2022 – Aug 2023

- Established the laboratory's initial hiPSC-derived cardiac organoid fabrication protocol, creating the foundation for subsequent organoid research.
- Fabricated C2C12-based 3D skeletal muscle tissues and applied cyclic mechanical stretch to assess tissue maturation.
- Joined as one of the laboratory's first three members and helped establish equipment, core experimental protocols, and safety procedures.

LABORATORY TOURS

Harvard University and Emory University, USA **Sep 2024**

- Visited the laboratories of Prof. Kit Parker (Harvard) and Prof. Sung Jin Park (Emory).

CONFERENCE PRESENTATIONS

- Poster; Korea BioChip Society (Nov 2025, Jeju) — Cardiac Organoid Platform with Micropillar-Assisted Calcium and Optical Signal Analysis
- Poster; 2025 Summer Symposium on Biological Convergence Engineering (Sep 2025, Sejong University) — Chamber-Specific Cardiac Organoid Platform with Micropillar-Assisted Calcium and Optical Signal Analysis

TEACHING AND MENTORING

Undergraduate Research Mentor, SURF Program Jul 2026 – Aug 2026

- Mentored 4 undergraduate researchers in cell culture, experimental planning, confocal imaging, and final project presentations.

Teaching Assistant, Biochemistry Laboratory Jul 2023 – Dec 2024

- Guided 120 students through antibody-production workflows, from plasmid transformation to protein purification and Western blotting; prepared teaching materials.

*Teaching Assistant***,** Microbiology Laboratory Jan 2024 – Jun 2024

- Instructed 40 students in bacterial culture, Gram staining, biochemical assays, and API-based identification.

*Peer Research Mentor***,** DoLearn-DoRun Program Sep 2023 – Dec 2023

- Mentored 2 undergraduates investigating myotube alignment on patterned substrates; authored the proposal and directed experiments.

SKILLS

Cell & Tissue Engineering: hiPSC culture; chamber-specific cardiomyocyte differentiation; 3D cardiac organoid and skeletal muscle tissue engineering.

Molecular & Cellular Assays: qRT-PCR; flow cytometry; immunofluorescence staining.

Imaging & Functional Analysis: confocal microscopy; Fluo-4–based calcium-transient imaging; calcium-based activation mapping; brightfield-based contractility analysis.

Design & Fabrication: Rhino; FDM and SLA 3D printing; laser cutting.

Programming & Computational Analysis: Python; MATLAB; COMSOL Multiphysics; ImageJ; GraphPad Prism.

Scientific Communication: Adobe Illustrator; EndNote.

REFERENCES

Keel Yong Lee, PhD

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Relationship: Master's thesis advisor and current research supervisor